In [1]:
```
import datetime
import time
```

Functions ¶

In [ ]:
```
1)time.time()
2)time.perf_counter()
3)time.sleep()
4)datetime.datetime(2022,3,24)

#Today Date and Time
5)datetime.datetime.now()
6)datetime.datetime.today()

#Todays Date
7)datetime.datetime.today().date()
8)datetime.datetime.now().date()

10)datetime.strftime()
11)datetime.datetime.strptime(date,date_format)
12)datetime.timedelta()
```

In [2]:
```
date1="02-1-1995"
date2="02 Nov 2022"

adhar_dob="11-02-1996"

pan_dob="11/02/1996"

type(adhar_dob)
```

Out[2]:
```
str
```

In [3]:
```
adhar_dob==pan_dob
```

Out[3]:
```
False
```

1.time.time()

In [4]:
```
Syntax:time.time()
```

In [5]:
```
time.time()
```

Out[5]:
```
1787049896.1204135
```

In [6]:
```
start_time=time.time()

for i in range(500000000):
    pass

end_time=time.time()
total_time=end_time-start_time
print("Total time:",total_time)
```

```
Total time: 28.237667560577393
```

2.time.perf_counter()

In [7]:
```
def KNN_Classifier():
    print("KNN classifier")
    for i in range(400000):
        pass
start_time=time.perf_counter()
KNN_Classifier()
end_time=time.perf_counter()

total_time=end_time-start_time
print("Total Time:",total_time)
```

```
KNN classifier
Total Time: 0.011278400000037436
```

3.time.sleep()

In [ ]:
```
Synatx:
    time.sleep(seconds)
```

In [8]:
```
print("Hello")
print("Data Science")
print("MAchine Learning")
print("Deep learning")
```

```
Hello
Data Science
MAchine Learning
Deep learning
```

In [9]:
```
print("Hello")
print("Data Science")
print("MAchine Learning")
time.sleep(5)
print("Deep learning")
```

```
Hello
Data Science
MAchine Learning
Deep learning
```

In [10]:
```
print("Hello")
print("Data Science")
time.sleep(5)
print("MAchine Learning")
time.sleep(5)
print("Deep learning")
```

```
Hello
Data Science
MAchine Learning
Deep learning
```

In [11]:
```
list1=["Pune","Mumbai","Delhi","Chennai","Kolkata"]
for city in list1:
    print("City Name is:",city)
    time.sleep(4)
```

```
City Name is: Pune
City Name is: Mumbai
City Name is: Delhi
City Name is: Chennai
City Name is: Kolkata
```

Datetime

In [12]:
```
import datetime
```

4)datetime.datetime()

1.Current Dattime

In [13]:
```
current_datetime=datetime.datetime.now()
print("Current Datetime:",current_datetime)
```

```
Current Datetime: 2026-08-18 16:28:51.921766
```

In [14]:
```
current_datetime=datetime.datetime.now()
print("Current Datetime:",current_datetime)
```

```
Current Datetime: 2026-08-18 16:29:16.353224
```

In [16]:
```
current_datetime=datetime.datetime.today()
print("Current Datetime:",current_datetime)
```

```
Current Datetime: 2026-08-18 16:30:13.043186
```

In [17]:
```
type(current_datetime)
```

Out[17]:
```
datetime.datetime
```

In [18]:
```
current_date=datetime.datetime.today().date()
print("Current Date:",current_date)
```

```
Current Date: 2026-08-18
```

In [19]:
```
current_date=datetime.datetime.now().date()
print("Current Date:",current_date)
```

```
Current Date: 2026-08-18
```

In [20]:
```
current_date1=datetime.datetime.today().date()
current_date2=datetime.datetime.now().date()
current_date3=datetime.date.today()

print(current_date1)
print(current_date2)
print(current_date3)
```

```
2026-08-18
2026-08-18
2026-08-18
```

Current Time

In [21]:
```
current_time=datetime.datetime.now().time()
print("Current Time:",current_time)
```

```
Current Time: 16:35:09.713365
```

In [22]:
```
current_time=datetime.datetime.today().time()
print("Current Time:",current_time)
```

```
Current Time: 16:35:31.337341
```

In [24]:
```
current_datetime=datetime.datetime.today()
print("Current_datetime:",current_datetime)
```

```
Current_datetime: 2026-08-18 16:37:05.866435
```

In [25]:
```
current_datetime.year
```

Out[25]: 2026

In [26]:
```
current_datetime.month
```

Out[26]: 8

In [27]:
```
current_datetime.day
```

Out[27]: 18

In [28]:
```
current_datetime.hour
```
Out[28]:
```
16
```

In [29]:
```
current_datetime.minute
```
Out[29]:
```
37
```

In [31]:
```
current_datetime.second
```
Out[31]:
```
5
```

In [32]:
```
current_datetime.microsecond
```
Out[32]:
```
866435
```

time Delta

In [ ]:
```
By using this we can find out past or future datetime.
```

In [ ]:
```
datetime.timedelta(days=0,seconds=0,microseconds=0,
                   miliseconds=0,minutes=0,hours=0,week=0)
 Return Date
```

Print prevoius 5 dates and last 5 dates from current dates

In [33]:
```
current_datetime=datetime.datetime.today().date()
for i in range(-5,6):
    new_date=current_datetime+datetime.timedelta(days=i)
    print("New_date:",new_date)
```
```
New_date: 2026-08-13
New_date: 2026-08-14
New_date: 2026-08-15
New_date: 2026-08-16
New_date: 2026-08-17
New_date: 2026-08-18
New_date: 2026-08-19
New_date: 2026-08-20
New_date: 2026-08-21
New_date: 2026-08-22
New_date: 2026-08-23
```

In [34]:
```
current_date=datetime.datetime.today().date()
new_date=current_date+datetime.timedelta(days=5)
print("New_date:",new_date)
```
```
New_date: 2026-08-23
```

In [35]:
```
current_date=datetime.datetime.today().date()
new_date=current_date+datetime.timedelta(days=10)
print("New_date:",new_date)
```

```
New_date: 2026-08-28
```

In [36]:
```
current_date=datetime.datetime.today().date()
new_date=current_date+datetime.timedelta(days=-7)
print("New_date:",new_date)
```

```
New_date: 2026-08-11
```

In [37]:
```
current_date=datetime.datetime.today().date()
new_date=current_date+datetime.timedelta(days=365)
print("New_date:",new_date)
```

```
New_date: 2027-08-18
```

In [43]:
```
current_date=datetime.datetime.today()
new_date=current_date+datetime.timedelta(hours=5)
print("New_date:",new_date)
```

```
New_date: 2026-08-18 21:52:09.458487
```

In [44]:
```
current_date=datetime.datetime.today()
new_date=current_date+datetime.timedelta(weeks=4)
print("New_date:",new_date)
```

```
New_date: 2026-09-15 16:52:21.162202
```

strftime

In [ ]:
```
strftime>> Strinf from datetime object
```

In [45]:
```
current_datetime=datetime.datetime.today()
print("Current datetime:",current_datetime)
```

```
Current datetime: 2026-08-18 16:54:45.978359
```

```
2026-08-18>>'18-08-2026'
2026-08-18>>'18/08/2026'
2026-08-18>>'18 Aug 2026'
2026-08-18>>'18 August 2026'
2026-08-18>>'2026-08-18'
2026-08-18>>'18-08-26'
2026-08-18>>18/08/26
```